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MODULE 9 HANDBOOK

# Module 9: AMPscript Programming

7 sections, 60 topics explained, plus the detailed lessons, examples, exercises and official links.

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Edition: November 2026. Practice labs marked "account" need a provisioned Marketing Cloud account. Other labs use sample files or paper.

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# Module 9: AMPscript Programming

**Level:** Intermediate. AMPscript: syntax, variables, personalisation, data extension lookups and debugging.

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# Detailed lessons

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## 9.1 Syntax, variables and personalisation

### WHAT IS IT?

> AMPscript runs on the server inside %%[ ... ]%% blocks and prints values with %%=...=%%. SET assigns a variable; AttributeValue reads a subscriber attribute; ProperCase formats names.

### WHY IS IT IMPORTANT?

> Almost every personalised SFMC email or CloudPage uses AMPscript.

### HOW DOES IT WORK AND THE STEPS

1. Declare with VAR and assign with SET.
2. Print with %%= ... =%%.
3. Use AttributeValue for subscriber fields and provide a default in IIF or a check.

### REAL-WORLD EXAMPLE

> Hello %%=ProperCase(AttributeValue("FirstName"))=%%.

### CODE EXAMPLE, WITH LINE-BY-LINE NOTES

```
%%[
  VAR @first, @greeting
  SET @first = AttributeValue("FirstName")
  IF Empty(@first) THEN
    SET @greeting = "there"
  ELSE
    SET @greeting = ProperCase(@first)
  ENDIF
]%%
Hello %%=v(@greeting)=%%,
// Line by line: VAR declares the variables; SET reads the subscriber's first name;
// IF/ELSE chooses the fallback when the field is empty; v() prints the variable.
```

### COMMON ERRORS

Printing an empty first name, which shows 'Hello ,'.

### INTERVIEW QUESTION

What does AttributeValue return if the attribute is missing?

### HANDS-ON EXERCISE

Write a greeting that falls back to 'there' if FirstName is empty.

### OFFICIAL DOCUMENTATION AND FURTHER READING

- AMPscript guide (Salesforce docs)

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### 9.2 Data extension lookups

#### WHAT IS IT?

> Lookup returns one value from a data extension by matching a field. LookupRows returns a set of rows. RowCount counts them. Always handle the case when nothing is found.

#### WHY IS IT IMPORTANT?

> Lookups personalise messages from other data (orders, loyalty points) without a separate send.

#### HOW DOES IT WORK AND THE STEPS

1. Use Lookup('DataExtension','ReturnField','MatchField',value).
2. Check RowCount on LookupRows before looping.
3. Use the subscriber key or email address as the match value.

#### REAL-WORLD EXAMPLE

> Show the loyalty points for the subscriber's email.

#### CODE EXAMPLE, WITH LINE-BY-LINE NOTES

```
%%[
  VAR @tier
  SET @tier = Lookup("LoyaltyMembers", "Tier", "EmailAddress", emailaddr)
  IF Empty(@tier) THEN SET @tier = "Member" ENDIF
]%%
Your tier: %%=v(@tier)=%%
// Lookup matches EmailAddress to the subscriber email (emailaddr); if no row, @tier is empty and
the default is used.
```

#### COMMON ERRORS

Lookup on a field that is not unique and returns a random row.

#### INTERVIEW QUESTION

Explain Lookup and LookupRows.

#### HANDS-ON EXERCISE

Show the loyalty tier with a Lookup and a default when no row exists.

#### OFFICIAL DOCUMENTATION AND FURTHER READING

- AMPscript guide (Salesforce docs)

## 9.1 AMPscript Fundamentals: every topic explained

**What is AMPscript?.** AMPscript is Salesforce Marketing Cloud's scripting language for personalising emails, SMS, CloudPages and landing pages at send or page-load time.

**AMPscript execution.** It runs on SFMC servers when the message is built or the page loads, not in the recipient's email client or browser.

**Syntax.** Code blocks use %%[ ... ]%%; inline output uses %%= ... =%%; functions take arguments in brackets.

**Variables.** Named values that start with @, such as @firstName.

**Functions.** Built-in operations such as Lookup, Concat, Format and ProperCase.

**Inline AMPscript.** Short expressions printed directly in content with %%= ... =%%.

**AMPscript blocks.** Multi-line logic between %%[ and ]%%, used to set variables and run conditions.

**Output expressions.** The printed result of an expression, such as %%=v(@firstName)=%%.

## 9.2 Variables and Data: every topic explained

**SET.** Assigns a value: SET @x = 1.

**SETROW.** Sets the row set returned by LookupRows into a variable for looping (used with Row and Field).

**Variable declaration.** VAR declares variables before use: VAR @a, @b.

**Data types.** AMPscript values are strings, numbers, dates, booleans or row sets.

**String variables.** Text values in double quotes.

**Numeric variables.** Numbers used in maths and comparisons.

**Date variables.** Dates used with DateAdd, DateDiff and FormatDate.

## 9.3 Personalisation: every topic explained

**AttributeValue.** Returns the value of a subscriber or sendable data extension field, or an empty value if missing.

**ProperCase.** Capitalises the first letter of each word.

**Uppercase.** Converts text to upper case.

**Lowercase.** Converts text to lower case.

**Substring.** Returns part of a string from a start position and length.

**Concat.** Joins strings together.

**Format functions.** Format and FormatDate change how numbers and dates display.

**Default values.** Use Empty() checks or IIF to provide a fallback value.

**Conditional personalisation.** Show different text based on data, such as tier or city.

## 9.4 Conditional Logic: every topic explained

**IF.** Runs code when a condition is true.

**ELSEIF.** Checks another condition when the first is false.

**ELSE.** Runs when no condition is true.

**FOR loops.** Repeats for a counter range, often over rows from LookupRows.

**WHILE loops.** Repeats while a condition holds. Use with care to avoid loops that never end.

**Conditional rendering.** Show content only when a condition is met.

**Dynamic content logic.** Choose content variants with IF logic based on subscriber data.

## 9.5 Data Extension Functions: every topic explained

**Lookup.** Returns one field value from the first row that matches.

**LookupRows.** Returns all rows that match one or more field values.

**LookupOrderedRows.** Returns matching rows sorted and limited by count.

**LookupRowsCS.** Case-sensitive version of LookupRows.

**LookupOrderedRowsCS.** Case-sensitive version of LookupOrderedRows.

**Row.** Gets a row by position from a row set.

**RowCount.** Counts rows in a row set; check it before looping.

**Field.** Reads a field from a row.

**InsertDE.** Inserts a row into a data extension (in emails, inserts at send time).

**UpdateDE.** Updates rows that match keys.

**UpsertDE.** Updates or inserts by key, often used on CloudPages.

**DeleteDE.** Deletes rows that match keys.

**RetrieveSalesforceObjects where supported.** Reads Salesforce CRM objects where the connector is set up and the context supports it.

## 9.6 Advanced AMPscript: every topic explained

**Error handling.** Guard every lookup and value with checks so a missing value does not break the message.

**Personalised URLs.** Build links with subscriber parameters, encoded safely.

**Dynamic images.** Choose image URLs from data, such as a product image.

**Dynamic content.** Show content blocks by rules or data.

**Date calculations.** DateAdd, DateDiff and FormatDate for expiry dates and reminders.

**Subscriber-specific content.** Content based on the subscriber's own data.

**Data extension lookups.** Reading related data such as orders or loyalty points at send time.

**Email personalisation.** Names, offers, products and dates personalised in email.

**CloudPages personalisation.** Reading request parameters and writing to data extensions on pages.

## 9.7 AMPscript Debugging: every topic explained

**Common errors.** Syntax errors, wrong function names, missing quotes and missing ENDIF.

**Invalid field references.** A field name that does not exist in the data source.

**Lookup errors.** Lookups on fields that are not unique, or data extensions with wrong names.

**Missing values.** Empty values when data is missing; always provide defaults.

**Syntax errors.** Unclosed blocks or brackets that stop the preview.

**Preview troubleshooting.** Preview with several test subscribers to find data problems.

**Performance considerations.** Many lookups per message slow sends; pre-join data with SQL where possible.